

Syllabus: COMP 364/464

High-Performance Computing

Credit Hours: 3

Prerequisites: COMP 413

Schedule: W 7:00PM - 9:30PM

Location: Zoom

Instructor: Melanie Cornelius

mcornelius@luc.edu

Office hours: By appointment, W 6:15-7:00 PM - [Calendly](#)

Course Description

Official Description: This course will use a blend of foundational understanding as well as a set of practical tools to gain insight into performance engineering of software. Students will learn shared memory, message passing and hybrid models of programming in both tightly-coupled and loosely-coupled computer systems

Grades

We will be using a standard four-point grading scale (see right).

Grades will be curved and are broken down according to:

- Exams:
 - Midterm: 25%
 - Final: 25%
 - Example exams (for format) and solutions (for expectations) will be provided at the review
 - Depending on curve, might modify this to 20-30 if it's advantageous to all students
- Projects:
 - P1, Systems survey: 10%
 - P2, Programming and Performance: 15%
 - P3, Capstone: 25%
 - Rubrics and examples will be posted

Letter	%	GP
A	95 – 100%	4.0
A-	90 – 95%	3.7
B+	87 – 90%	3.3
B	84 – 87%	3.0
B-	80 – 84%	2.7
C+	77 – 80%	2.3
C	74 – 77%	2.0
C-	70 – 74%	1.7
D+	67 – 70%	1.3
D	64 – 67%	1.0
D-	60 – 64%	0.7
F	< 60%	0.0

Course Weekly Schedule and Structure

This schedule is subject to change; changes will be announced in-class and on Piazza. See the below charts for a description of topics, weeks, and project deadlines.

This course is remote and will meet using Zoom on Wednesdays, from 7:00 - 9:30 PM. Recordings are taken of each class but are *not* guaranteed; students should plan to attend synchronously.

Office Hours: Virtual office hours will be held for the 45 minutes before class, Wednesdays 6:15 - 7:00 PM.

Appointments can also be made as-needed. The instructor's schedule is posted to [Calendly](#). Please provide at least 24 hours notice for appointments outside the regularly-scheduled time.

Course Weekly Schedule

Wk	Date	Topic	Readings	HW	Projects
1	Aug 26	Intro to the class + planning questionnaire	-	-	-
2	Sep 2	Intro to HPC: architectures, networks, topics overview	-	-	
3	Sep 9	Networks + architectures (cont.) + Linux, finalize	R1		Project 1 Announced
4	Sep 16	Parallel programming (Intro)	R2		
5	Sep 23	Parallel programming (OpenMP)	R3	HW 05	
6	Sep 30	Parallel programming (MPI)	R4	HW 06	Project 1 Due, 2 and 3 Announced
7	Oct 7	Performance monitoring + midterm review	R5	HW 07	
8	Oct 14	Midterm	-		
9	Oct 21	Virtualization in HPC	R6		Project 2 due, Project 3 plan due
10	Oct 28	Data in HPC + Security	R7		
11	Nov 4	Fault tolerance	R8		
12	Nov 11	AI in HPC	R9		
13	Nov 18	Open: possible guest speaker	R10		Project work time
14	Nov 25	Thanksgiving break	R11 (opt)	-	
15	Dec 2	Wrap-up / presentations + final review	-	-	Project 3 due

Project	Part	Description	Assigned	Due
1	-	Survey of networks and architectures in modern HPC	Wk 3 (Sep 9)	Wk 6 (Sep 30)
2	A	Parallel programming	Wk 6 (Sep 30)	Wk 9 (Oct 21)
	B	Performance Analysis	Wk 6 (Sep 30)	Wk 9 (Oct 21)
3	-	Topic selection (in your area)	Wk 6 (Sep 30)	Wk 9 (Oct 21)
3	A	Survey on your topic (draft Wk 8, final Wk 15)	Wk 6 (Sep 30)	Wk 15 (Dec 2)
3	B	Programming assignment on your topic	Wk 6 (Sep 30)	Wk 15 (Dec 2)
3	C	Presentation (10 to 20 min, per-recorded OK; live = extra credit)	Wk 6 (Sep 30)	Wk 15 (Dec 2)

Exams: We will have two exams during the semester, one midterm and one final. The format of the exams will be reviewed in-class. The exams will be in-person barring extenuating circumstances discussed well-ahead with the instructor.

364 vs 464 Expectations: Each project will have different requirements depending on the section in which the student is enrolled.

Breaks: We will not have assigned material during breaks in this course. For the Fall semester, this includes Thanksgiving break.

Textbook and Readings: We will occasionally reference M. van Steen and A.S. Tanenbaum, Distributed Systems, 4th ed., distributed-systems.net, 2023. This book is available for free from the author on [their website](#). Otherwise, readings will be posted the week before lectures to Piazza and will contain links to materials.

Completion of readings is not graded but is highly recommended. This course has a wide topic range to cover quickly, so familiarity with terms and the basics of topics will allow lectures to go into more interesting and complex material. In short, if a student doesn't do the readings, the pace of the associated lecture might be overly rapid. Do the readings!

Communication: We will be making use of Piazza for discussions in this course. You should have received an email inviting you to the COMP 150 Piazza page - please reach out to the instructor if not. This is the preferred mode of communication for any course-related questions or concerns.

Late Work: The policy on late work is fairly limited. In the case of extenuating circumstances, students should feel free to reach out to the instructor. Otherwise, each day an assignment is late will reduce the overall grade by 50%.

Since this is a 300-400 level course, and in the interest of independent responsibility, all students are permitted three no-questions-asked "late-credits". When used, each late-credit applies a 24-hour extension. These cannot be used on exams, but they can be freely applied to other deadlines unless specifically otherwise stated. Late-credits will be tracked using the gradebook on Sakai, and students wishing to use one of their late-credits need only inform the instructor (no explanation needed) *prior to submission*. Please also indicate on the assignment submission that a late-credit was used.

Policies

Academic Honesty: This class has a zero-tolerance policy on academic dishonesty. All material submitted must be the individual's own work. Labs, quizzes, exams, and discussions that are submitted and determined to not be the individual work of the submitter will be assigned a zero grade. University policy requires academic dishonesty to be reported to the university.

This is a serious matter: please, do not submit work that is not your own.

It is easy to catch, it will net a zero grade, and it has to be reported to the university - but more than that, doing so is disrespectful to yourself, the instructor, and to the university alike. It's not worth it.

Outside sources, such as other students, online answers, or material generated by LLMs (AI), are not considered individual work. You are welcome and encouraged to use an AI to assist your work, but documentation of the AI use must be included with any assignment. Further discussion on this topic will be explored in class. For any questions, please contact the instructor.

Student Support

Students are urged to contact the instructor should they have questions concerning course materials and procedures. If you have any special circumstance that may have some impact on your course work, please let me know so we can establish a plan for assignment completion. If you require assignment accommodations, please contact me early in the semester so that arrangements can be made with the Student Accessibility Center: www.luc.edu/sac/

Under Title IX, a federal civil rights law, Loyola is obligated to ensure a safe and inclusive academic environment free from sex-based discrimination. At Loyola, the Office for Equity & Compliance ("OEC") is the University's Title IX office and coordinates the University's response to reports and complaints of sexual misconduct (as well as discrimination of any kind) to ensure all students' rights are protected.

Links to Resources:

- Sakai: Student resource page for Sakai: <https://loyola.screenstepslive.com/>
- Academic calendar: https://www.luc.edu/academics/schedules/fall/academic_calendar
- Final Exam Schedules: https://www.luc.edu/academics/schedules/spring/exam_schedule
- Wellness center: <https://www.luc.edu/wellness/>
- Special accommodation and policies: Student Accessibility Center: <https://www.luc.edu/sac/>
- Student Support Services: <https://www.luc.edu/online/resources/studentssupportservices/>
- Tutoring Center: <https://www.luc.edu/tutoring>
- Notice of Reporting Obligations and Student Title IX Rights
- Religious Holiday Observance Attendance Statement: <https://www.luc.edu/academicaffairs/stories/archive/honoringstudentreligiousobservances.shtm>